

M54 Final Prep

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1 Introduction

Approach to solving final problems modeled based on Arun and Serganova's class material.

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- a. Show that T is a linear transformation.
 - b. Describe which vectors are in $\text{Ker } T$.
 - c. Find a basis for $\text{Ran } T$.
 - d. Let A be the matrix of T . Explain why A is diagonalizable, and find a matrix D for which there is an invertible matrix P with $A = PDP^{-1}$.
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- a. Show $T(0) = 0$, $T(u + v) = T(u) + T(v)$, and $T(cu) = cT(u)$
 - b. With this formula, $\dim(V) = \text{rank}(V) + \text{null}(A)$, we can describe which vectors are in $\text{Ker } T$.
 - c. Finding a basis for range of T —
 - d. For A to be diagonalizable it must not have any 0's on the D . However, one side effect of upper triangularized (partial rref) on a dependent matrix, is that it stays dependent (idiomatic, static) . Independent matrices can mutate

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- a. True, note that I think even if our eigenvalues are equal we still can express it orthogonal too.
- b. True, need to recite definition of similar matrices. Similar $\iff A = PBP^{-1} \wedge B = PAP^{-1}$

- c. True, need to recite that $D^T = D$, which we show this by having our definition that if A diagonalizable $\implies$ invertible meaning $A = PDP^{-1}$ meaning we can transpose A to show $D^T = D$.

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- a. $\forall A \in \mathbb{R}^{n \times n} : \langle \mathbf{x}, \mathbf{y} \rangle = (A\mathbf{x}) \cdot (A\mathbf{y})$ is an inner product on $\mathbb{R}^n$ - not for all, we have the condition that A must be independent
- b. If A orthogonal, then linear map $T(\mathbf{x}) = A\mathbf{x}$ is bijective
- c.

Proofs

Show $Nul(A^T A) = Nul(A)$

(1) **Def:** $Nul(A) := Ax = 0$, the homogeneous solution

$$Nul(A^T A) = Nul(A)$$

Apply (1) to show $Nul(A) \subseteq Nul(A^T A)$

$$A^T A = 0$$

Multiply both sides by A^T

$$A^T(A) = A^T(0)$$

Problem 3: Orthogonal Projection Matrix

Definition 1.1: Orthogonal Projection Matrix

Let $V \subseteq \mathbb{R}^{20}$ be a 4-dimensional subspace. The standard matrix A of $T(\mathbf{y}) = \text{proj}_V \mathbf{y}$ is a 20×20 matrix satisfying:

$$A^2 = A \quad (\text{idempotent}) \quad A^T = A \quad (\text{symmetric})$$

Key Fact 1.1: Eigenvalues of a Projection Matrix

From $A^2 = A$:

$$A^2 \mathbf{x} = A\mathbf{x} \implies \lambda^2 \mathbf{x} = \lambda \mathbf{x} \implies \lambda(\lambda - 1) = 0 \implies \lambda \in \{0, 1\}$$

(a) Eigenvalues of A

$$\lambda = 1 \text{ with multiplicity } 4, \quad \lambda = 0 \text{ with multiplicity } 16$$

Reasoning: $\lambda = 1$ corresponds to vectors *in* V (fixed by projection); $\lambda = 0$ corresponds to vectors in $V^\perp$ (annihilated by projection).

(b) Characteristic Polynomial

A is 20×20 , so $\det(A - \lambda I)$ is degree 20:

$$p(\lambda) = \lambda^{16}(\lambda - 1)^4$$

(c) Rank of A

$$\text{rank}(A) = \dim(V) = 4$$

The image of a projection onto V is exactly V , so $\text{Col}(A) = V$.

(d) Dimension of Nul A

By Rank-Nullity (A has $n = 20$ columns):

$$\text{nullity}(A) = 20 - \text{rank}(A) = 20 - 4 = 16$$

Alternatively: $\text{Nul}(A) = V^\perp$, which has $\dim(V^\perp) = 20 - 4 = 16$.

(e) Is A Symmetric?

Yes. By definition of orthogonal projection:

$$\langle A\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A\mathbf{y} \rangle \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^{20} \implies A^T = A$$